

		-10 W
28	A	Kinetic energy of electron, $E = \frac{1}{2}mv^2 \text{ --- (1)}$ Momentum of electron, $p = mv \text{ --- (2)}$ Combining both (1) and (2), $p = \sqrt{2mE}$ De Broglie wavelength of electron, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$ $\lambda \propto \frac{1}{\sqrt{E}}$ $\frac{\lambda'}{\lambda} = \sqrt{\frac{E}{9E}}$ $\therefore \lambda' = \frac{1}{3}\lambda$
29	C	Nuclear mass defect